

Homework 1

Due: Oct. 15 2026

1. Do Exercise 1 on page 7.
2. Consider the following boundary-value problem (i.e. strong-form problem)

$$(S) \begin{cases} u_{,xx} + f = 0, & \text{in } (0,1) \\ u(0) = g, \\ u_{,x}(1) = h. \end{cases}$$

Different from the problem used in our lecture, we here impose the Dirichlet (Neumann) boundary condition at $x=0$ (1), respectively.

- a. Construct its corresponding weak-form problem.
 - b. Show the equivalence between the strong- and weak-form problems.
3. Consider the boundary-value problem

$$\begin{aligned} u_{,xx} + f &= 0, \\ u(1) &= g, \\ -u_{,x}(0) &= h. \end{aligned}$$

Assume that $f = \sin(x)$.

- a. Show the following identities

$$(N_A, N_{A-1}) = \frac{h_{A-1}}{6}, \quad (N_A, N_A) = \frac{h_{A-1} + h_A}{3}, \quad (N_A, N_{A+1}) = \frac{h_A}{6}.$$

- b. Employing the linear finite element space with equally spaced nodes, set up the stiffness matrix and load vector using 3 elements. Solve the matrix problem. Use $\hat{f} \approx \sum_{B=1}^{n+1} f_B N_B$ in the assembly of the load vector. Do you get nodally exact solution here?
- c. Repeat the calculation without invoking the approximation $\hat{f}$. In other words, calculate (N_A, f) rather than $(N_A, \hat{f})$ in the load vector. What do you observe? Notice that the following identity can be helpful.

$$\int x \sin(x) dx = - \int x d\cos(x) = -x\cos(x) + \int \cos(x) dx = -x\cos(x) + \sin(x).$$